

System of Linear Equations

A collection of one or more linear equations with same number of unknowns $x_1, \dots, x_n$

e.g. $2x_1 - x_2 + 1.5x_3 = 8$ ①
 $x_1 - 4x_3 = -7$ ②

solutions: $(x_1 = -7, x_2 = -22, x_3 = 0)$, $(x_1 = 1, x_2 = -3, x_3 = 2)$,

solution set:

$$\textcircled{1} - 2 \times \textcircled{2} \Rightarrow -x_2 + 1.5x_3 - (-8x_3) = 8 - (-14)$$

$$\Rightarrow -x_2 + 9.5x_3 = 22$$

$$\Rightarrow x_2 = 9.5x_3 - 22$$

$$\textcircled{2} \Rightarrow x_1 = 4x_3 - 7$$

Then the solution set is the set of (x_1, x_2, x_3) such that

$x_1 = 4x_3 - 7$ and $x_2 = 9.5x_3 - 22 \Rightarrow$ infinitely many solutions

A system of equations can have

{	1. no solution	inconsistent
	2. exactly one solution	consistent
	3. infinitely many solution	

Row Reduction

$$\begin{cases} X_1 + 2X_2 - X_3 = 1 \\ -X_1 + X_3 = -2 \\ X_2 + 3X_3 = 0 \end{cases} \longrightarrow \begin{array}{cccc} & X_1 & X_2 & X_3 & \text{constant} \\ \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ -1 & 0 & 1 & -2 \\ 0 & 1 & 3 & 0 \end{array} \right] \end{array}$$

↑ augmented column.

equations = # rows

unknowns (variables) = # columns.

Definition. A matrix is in *row echelon form* if:

1. All zero rows are at the bottom.
2. The first nonzero entry of a row is to the *right* of the first nonzero entry of the row above.
3. Below the first nonzero entry of a row, all entries are zero.

Here is a picture of a matrix in row echelon form:

$$\begin{pmatrix} \boxed{\star} & \star & \star & \star & \star \\ 0 & \boxed{\star} & \star & \star & \star \\ 0 & 0 & 0 & \boxed{\star} & \star \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$\star$ = any number
 $\boxed{\star}$ = any nonzero number

Definition. A **pivot** is the first nonzero entry of a row of a matrix in row echelon form.

Definition. A matrix is in **reduced row echelon form** if it is in row echelon form, and in addition:

4. Each pivot is equal to 1.
5. Each pivot is the only nonzero entry in its column.

$$\begin{pmatrix} \boxed{\star} & \star & \star & \star & \star \\ 0 & \boxed{\star} & \star & \star & \star \\ 0 & 0 & 0 & \boxed{\star} & \star \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$\star$ = any number

$\boxed{\star}$ = any nonzero number

REF

$$\begin{pmatrix} \color{red}{1} & 0 & \star & 0 & \star \\ 0 & \color{red}{1} & \star & 0 & \star \\ 0 & 0 & 0 & \color{red}{1} & \star \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$\star$ = any number

$\color{red}{1}$ = pivot

RREF

Row Operations you can do

- Scaling (e.g. $R_1 \rightarrow 2R_1$)
- Replacement (e.g. $R_2 \rightarrow R_2 - R_1$)
- Swap (e.g. $R_1 \leftrightarrow R_2$)

Row Reduction Recipe

1. Start at the top-left non-zero entry, make it 1.
2. Clear down.
3. Go to the next non-zero entry following the "downstair" direction.
4. Repeat until REF
5. Clear up until RREF

Get a 1 here

$$\begin{pmatrix} \boxed{1} & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{pmatrix}$$

Clear down

$$\begin{pmatrix} \boxed{1} & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{pmatrix}$$

Get a 1 here

$$\begin{pmatrix} 1 & * & * & * \\ 0 & \boxed{1} & * & * \\ 0 & * & * & * \\ 0 & * & * & * \end{pmatrix}$$

Clear down

$$\begin{pmatrix} 1 & * & * & * \\ 0 & \boxed{1} & * & * \\ 0 & * & * & * \\ 0 & * & * & * \end{pmatrix}$$

(maybe these are already zero)

$$\begin{pmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & \boxed{0} & * \\ 0 & 0 & 0 & * \end{pmatrix}$$

Get a 1 here

$$\begin{pmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & \boxed{*} \\ 0 & 0 & 0 & * \end{pmatrix}$$

Clear down

$$\begin{pmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & * \end{pmatrix}$$

Matrix is in REF

$$\begin{pmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Clear up

$$\begin{pmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Clear up

$$\begin{pmatrix} 1 & * & * & 0 \\ 0 & \boxed{1} & * & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$x_1 \ x_2 \ x_3 \ x_4$
Matrix is in RREF

$$\begin{pmatrix} 1 & 0 & * & 0 \\ 0 & 1 & * & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

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* = any number

$\boxed{*}$ = any nonzero number

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$\boxed{\star}$ = any nonzero number

REF

$$\begin{pmatrix} \color{red}{1} & 0 & \star & 0 & \star \\ 0 & \color{red}{1} & \star & 0 & \star \\ 0 & 0 & 0 & \color{red}{1} & \star \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$\star$ = any number

$\color{red}{1}$ = pivot

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Get a 1 here

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Clear down

$$\begin{pmatrix} \boxed{1} & * & * & * \\ \downarrow & * & * & * \\ * & * & * & * \\ * & * & * & * \end{pmatrix}$$

Get a 1 here

$$\begin{pmatrix} 1 & * & * & * \\ 0 & \boxed{1} & * & * \\ 0 & * & * & * \\ 0 & * & * & * \end{pmatrix}$$

Clear down

$$\begin{pmatrix} 1 & * & * & * \\ 0 & \boxed{1} & * & * \\ \downarrow & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \end{pmatrix}$$

(maybe these are already zero)

$$\begin{pmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & \boxed{0} & * \\ 0 & 0 & 0 & * \end{pmatrix}$$

Get a 1 here

$$\begin{pmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & \boxed{*} \\ 0 & 0 & 0 & * \end{pmatrix}$$

Clear down

$$\begin{pmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & * \end{pmatrix}$$

Matrix is in REF

$$\begin{pmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Clear up

$$\begin{pmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Clear up

$$\begin{pmatrix} 1 & * & * & 0 \\ 0 & \boxed{1} & * & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$x_1 \ x_2 \ x_3 \ x_4$
Matrix is in RREF

$$\begin{pmatrix} 1 & 0 & * & 0 \\ 0 & 1 & * & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Parametric Form

Definition. Consider a *consistent* system of equations in the variables $x_1, x_2, \dots, x_n$. Let A be a row echelon form of the augmented matrix for this system.

We say that x_i is a **free variable** if its corresponding column in A is *not* a pivot column.

Reduce to REF/RREF $\Rightarrow$ determine pivot columns & free variables.
 $\Rightarrow$ write parametric form.

Recipe: Parametric form.

The **parametric form** of the solution set of a consistent system of linear equations is obtained as follows.

1. Write the system as an augmented matrix.
2. Row reduce to reduced row echelon form.
3. Write the corresponding (solved) system of linear equations.
4. Move all free variables to the right hand side of the equations.

e.g. $\begin{cases} 2x + y + 12z = 1 \\ x + 2y + 9z = -1 \end{cases}$ $\left[\begin{array}{ccc|c} 2 & 1 & 12 & 1 \\ 1 & 2 & 9 & -1 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 0 & 5 & 1 \\ 0 & 1 & 2 & -1 \end{array} \right]$

$\begin{cases} x_1 + 5x_3 = 1 \\ x_2 + 2x_3 = -1 \end{cases} \Rightarrow \begin{cases} x_1 = 1 - 5x_3 \\ x_2 = -1 - 2x_3 \end{cases}$ $x = \begin{bmatrix} -5 \\ -2 \\ 1 \end{bmatrix} x_3 + \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ $\leftarrow$ parametric form.

Note: In the RREF matrix, columns 1 and 2 are pivots, and column 3 is a free variable.

Vectors

Vector addition and scalar multiplication.

- We can add two vectors together:

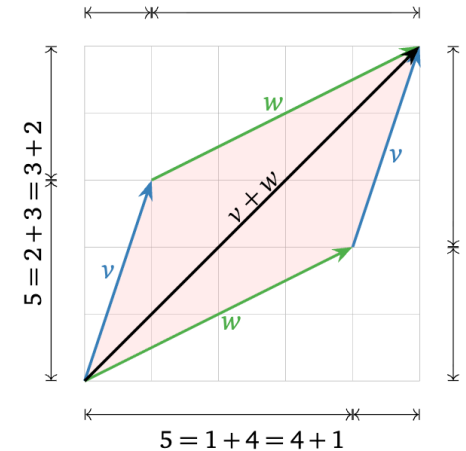
$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} + \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a+x \\ b+y \\ c+z \end{pmatrix}.$$

- We can multiply, or **scale**, a vector by a real number c :

$$c \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} c \cdot x \\ c \cdot y \\ c \cdot z \end{pmatrix}.$$

We call c a **scalar** to distinguish it from a vector. If v is a vector and c is a scalar, then cv is called a **scalar multiple of v** .

Addition and scalar multiplication work in the same way for vectors of length n .



The parallelogram law for vector addition.

Linear Combinations

Definition. Let $c_1, c_2, \dots, c_k$ be scalars, and let $v_1, v_2, \dots, v_k$ be vectors in $\mathbb{R}^n$. The vector in $\mathbb{R}^n$

$$c_1 v_1 + c_2 v_2 + \dots + c_k v_k$$

is called a **linear combination** of the vectors $v_1, v_2, \dots, v_k$, with **weights** or **coefficients** $c_1, c_2, \dots, c_k$.

Vector Equations and Spans

Definition. A **vector equation** is an equation involving a linear combination of vectors with possibly unknown coefficients.

Asking whether or not a vector equation has a **solution** is the same as asking if a given vector is a **linear combination** of some other given vectors.

$$\text{e.g. } x \begin{bmatrix} 1 \\ 2 \\ 6 \end{bmatrix} + y \begin{bmatrix} -1 \\ -2 \\ -1 \end{bmatrix} = \begin{bmatrix} 8 \\ 16 \\ 3 \end{bmatrix}$$

Span

! Definition. Let $v_1, v_2, \dots, v_k$ be vectors in $\mathbf{R}^n$. The **span** of $v_1, v_2, \dots, v_k$ is the collection of **all linear combinations** of $v_1, v_2, \dots, v_k$, and is denoted $\text{Span}\{v_1, v_2, \dots, v_k\}$. In symbols:

$$\text{Span}\{v_1, v_2, \dots, v_k\} = \{x_1 v_1 + x_2 v_2 + \dots + x_k v_k \mid x_1, x_2, \dots, x_k \text{ in } \mathbf{R}\}$$

We also say that $\text{Span}\{v_1, v_2, \dots, v_k\}$ is the subset **spanned by** or **generated by** the vectors $v_1, v_2, \dots, v_k$.

Matrix Equation

A is an $m \times n$ matrix
 $\uparrow$ $\uparrow$
 $\#$ rows $\#$ columns

$$A \vec{x} = \vec{b}$$

$\uparrow$ $\uparrow$ $\uparrow$
 $m \times n$ $n \times 1$ $m \times 1$

Understand $A\vec{x}$ by a linear combination of column vectors

↳ The **product** of A with a vector x in $\mathbb{R}^n$ is the linear combination

$$Ax = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = x_1 v_1 + x_2 v_2 + \cdots + x_n v_n.$$

Matrix multiplication recipe

↳ **Recipe: The row-column rule for matrix-vector multiplication.** If A is an $m \times n$ matrix with rows $r_1, r_2, \dots, r_m$, and x is a vector in $\mathbb{R}^n$, then

$$Ax = \begin{pmatrix} -r_1- \\ -r_2- \\ \vdots \\ -r_m- \end{pmatrix} x = \begin{pmatrix} r_1 x \\ r_2 x \\ \vdots \\ r_m x \end{pmatrix}.$$

Four Ways of Writing a Linear System. We now have *four* equivalent ways of writing (and thinking about) a system of linear equations:

1. As a system of equations:

$$\begin{cases} 2x_1 + 3x_2 - 2x_3 = 7 \\ x_1 - x_2 - 3x_3 = 5 \end{cases}$$

2. As an augmented matrix:

$$\left(\begin{array}{ccc|c} 2 & 3 & -2 & 7 \\ 1 & -1 & -3 & 5 \end{array} \right)$$

3. As a vector equation ($x_1 v_1 + x_2 v_2 + \cdots + x_n v_n = b$):

$$x_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + x_2 \begin{pmatrix} 3 \\ -1 \end{pmatrix} + x_3 \begin{pmatrix} -2 \\ -3 \end{pmatrix} = \begin{pmatrix} 7 \\ 5 \end{pmatrix}$$

4. As a matrix equation ($Ax = b$):

$$\begin{pmatrix} 2 & 3 & -2 \\ 1 & -1 & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 7 \\ 5 \end{pmatrix}.$$

In particular, all four have the same solution set.

Spans and Consistency.

The matrix equation $Ax = b$ has a solution if and only if b is in the span of the columns of A .

Theorem. Let A be an $m \times n$ (non-augmented) matrix. The following are equivalent:

1. $Ax = b$ has a solution for all b in $\mathbb{R}^m$.
2. The span of the columns of A is all of $\mathbb{R}^m$.
3. A has a pivot position in every row.

Solution Sets

Definition. A system of linear equations of the form $Ax = 0$ is called **homogeneous**.

A system of linear equations of the form $Ax = b$ for $b \neq 0$ is called **inhomogeneous**.

Observation. The equation $Ax = 0$ has a **nontrivial solution** $\iff$ there is a free variable $\iff A$ has a column without a pivot position.

$Ax=0$ has only **trivial solution** $\iff$ pivot in every column of A

The set of solutions to a **homogeneous** equation $Ax = 0$ is a span.

If $Ax = b$ is consistent, the set of solutions is obtained by taking one **particular solution** p of $Ax = b$, and adding all solutions of $Ax = 0$.

In particular, if $Ax = b$ is consistent, the solution set is a **translate of a span**.

Solution set can be written out using parametric form.

0 free variable	$\iff$	Solution set is a point
1 free variable	$\iff$	Solution set is a line
2 free variables	$\iff$	Solution set is a plane
3 free variables	$\iff$	Solution set is a space

Solution Sets and Column Spans * permalink

To every $m \times n$ matrix A , we have now associated two completely different geometric objects, both described using spans.

- The **solution set**: for fixed b , this is the set of all x such that $Ax = b$.
 - This is a **span** if $b = 0$, and it is a **translate of a span** if $b \neq 0$ (and $Ax = b$ is consistent).
 - It is a subset of $\mathbb{R}^n$.
 - It is computed by solving a system of equations: usually by row reducing and finding the parametric vector form.
- The **span of the columns of A** : this is the set of all b such that $Ax = b$ is consistent.
 - This is **always a span**.
 - It is a subset of $\mathbb{R}^m$.
 - It is not computed by solving a system of equations: row reduction plays no role.